Highlights

Real-time Implementation of Generalized Error Innovation-based Adaptive Estimation Kalman Filter for MIMO Active Noise Control

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- Address the problems faced by the LMS and standard Kalman filter (SKF) control in ANC.
- Propose an adaptive Kalman filtering algorithm to eliminate the need for proper initialization.
- Validate that the proposed algorithm performs similarly to SKF and facilitate practical deployment in real scenarios.
- Compare the proposed algorithm with other available adaptive Kalman filters.

Real-time Implementation of Generalized Error Innovation-based Adaptive Estimation Kalman Filter for MIMO Active Noise Control

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ABSTRACT

This paper proposes a generalized error innovation-based adaptive estimation Kalman filter (GeIAEKF) for multiple-input multiple-output (MIMO) active noise control (ANC) systems employing filtered-x structures. Standard Kalman filter (SKF)-based ANC methods offer strong tracking capabilities but rely on fixed, manually tuned covariance matrices, limiting their applicability in real-time and dynamic acoustic environments. The proposed GeIAEKF incorporates an error-driven adaptive mechanism that updates both the process and measurement noise covariance matrices online using only the measured residual error signal. Simulation results demonstrate that the GeIAEKF achieves a convergence speed comparable to the SKF while maintaining steady-state attenuation within 2-5 dB of the SKF performance across both stationary and non-stationary scenarios. Under gradual dynamic conditions, the proposed method exhibits fast adaptation and improved transient stability compared to some available adaptive Kalman filters. Real-time experiments further confirm its practical suitability for embedded ANC systems. The proposed GeIAEKF therefore bridges the gap between high-performance state-estimation-based ANC and practical real-time implementation in dynamic MIMO environments.

1. Introduction

Active noise control (ANC) attenuates unwanted sound by generating anti-noise through secondary actuators (Tsuji, Toyooka and Kajikawa, 2025; Zhang, Han and Duan, 2026). Advances in signal processing and embedded computation have expanded ANC applications to automotive cabins (Diaz, Egaña and Vinolas, 2006; Jung, Elliott and Cheer, 2019; Zhao, Chen, Zhou, Xu and Jia, 2024; Zheng, Wan, Jia, Li, Liu, Qiu and Liu, 2024), aircraft interiors (Haase, Unruh, Algermissen and Pohl, 2016), duct systems (Eriksson, 1991), high-speed trains (Diaz et al., 2006), and architectural acoustics (Murao, Shi, Gan and Nishimura, 2017).

At the core of an ANC system lies an adaptive control filter that identifies unknown or time-varying acoustic paths in real time (Widrow and Walach, 1983). However, in practical implementations, the presence of a secondary acoustic path between the control actuator and the error sensor prevents the direct use of conventional adaptive algorithms (Elliot, 2001). This limitation is typically addressed through the use of filtered-x structures (Douglas, 2002), which form the foundation of most modern ANC algorithms.

Adaptive ANC methods are commonly evaluated based on computational complexity, convergence speed, steady-state error, and tracking capability (Shi, Lam, Gan and Wen, 2021; Lian, Li, Sun, Wang and Lu, 2025). The filtered-x least mean squares (FxLMS) algorithm remains the most widely adopted solution due to its simplicity and low computational cost (Borchi, Carfagni, Martelli, Turchi and Argenti, 2016; Zhang, Zhang, Meng and Pi, 2023; Li, Lu, Chen, Zhu, Liu, Cheng and Sun, 2025; Jia, Chen, Zhao and Qian, 2025). However, LMS-based approaches exhibit slow convergence under dynamic disturbances (Cheer and Elliott, 2015; Liebich, Jax and Vary, 2016), limited robustness to impulsive or colored noise (Liu, Du and Wu, 2025; Luo, Chen, Yang, Zhang, Tian, Wang and Qiao, 2026), and performance degradation in multi-reference systems caused by eigenvalue spread (Zhang, Wang, Guo, Yang, Wang and

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Liu, 2019; Wang, He, Xia, Zhang and Xu, 2024a). Moreover, the use of a fixed step size introduces an inherent trade-off between adaptation speed and steady-state error (Bjarnason, 1992; Ferrer, de Diego, García-Mollá and Gonzalez, 2023). These limitations motivate the exploration of state-estimation-based adaptive control frameworks.

The Kalman filter (KF) provides a statistically optimal recursive estimation framework under linear-Gaussian assumptions and is well known for its superior tracking capability in non-stationary environments. KF-based ANC formulations have demonstrated improved convergence and disturbance tracking compared to LMS-based methods (Lopes and Piedade, 2000; Van Ophem and Berkhoff, 2013; Fabry, Liebich, Vary and Jax, 2018; Gowtham, Burra, Kar, Østergaard, Sooraksa, Mladenovic and Haddad, 2022). Nevertheless, practical implementation of KF-based ANC remains constrained by two critical limitations: its high computational complexity and the strong sensitivity of filter performance to the accurate specification of the process and measurement noise covariance matrices, $\mathbf{Q}(n)$ and $\mathbf{R}(n)$ (Kalman, 1960; Brown and Hwang, 1992).

In ANC applications, reliable prior knowledge of these covariance matrices is rarely available, and inaccurate initialization may lead to biased estimates, slow adaptation, or even filter divergence. However, most implementations rely on manually tuned, fixed covariance matrices, which undermines robustness in time-varying acoustic environments (Zhao and Huang, 2020). Adaptive or parameter-tuning Kalman filtering approaches have been proposed to reduce dependence on accurately initialized fixed covariance matrices. Although such methods have been extensively investigated in fields such as target tracking (Zhong, Peng, He, Feng, Li and Wang, 2024), dynamic state estimation in electrical machines (Ding, 2014), and lithium-ion battery monitoring (Wang, Dang, Gao, Li, Fernandez and Blaabjerg, 2024b), their application to MIMO filtered-x ANC remains limited. Existing adaptive ANC formulations (Liebich, Fabry, Jax and Vary, 2017) jointly adapt $\mathbf{Q}(n)$, $\mathbf{R}(n)$, and the state transition matrix $\mathbf{F}(n)$, increasing computational complexity and introducing structural inconsistency with the ANC objective.

Therefore, despite decades of research on KF-based ANC, no computationally efficient adaptive Kalman framework specifically tailored for MIMO filtered-x structures has yet been established, one that simultaneously eliminates the need for accurate prior covariance knowledge, reduces computational burden, and remains suitable for real-time implementation.

To address this open problem, this paper introduces a generalized error innovation-based adaptive estimation Kalman filter (GeIAEKF) specifically designed for MIMO filtered-x ANC systems. Unlike conventional innovation-based adaptive estimation Kalman filters (IAEKF) (Jin, Guo, He and Guo, 2017; Silva, Frutuoso, Souza and de Barros, 2022), which rely on innovation statistics for covariance adaptation, the proposed method directly exploits the measured error signal to adapt $\mathbf{Q}(n)$ and $\mathbf{R}(n)$ online. This ensures that covariance adaptation remains fully consistent with the ANC structure while eliminating the need for accurate prior covariance initialization. Existing Kalman-based ANC methods provide strong performance but are often impractical due to covariance tuning and computational burden; this work overcomes both limitations while preserving near-optimal performance.

The main contributions of this work are as follows:

- An online adaptive Kalman filtering scheme for MIMO filtered-x ANC in which covariance adaptation is explicitly derived from the measured error signal.
- Derivation of a generalized error innovation-based adaptive estimation Kalman filter (GeIAEKF) featuring analytically simplified update expressions for the Kalman gain and state-error covariance.
- Thorough simulation studies demonstrate convergence speed, steady-state mean squared error, and tracking performance under both stationary and dynamic conditions; the proposed algorithm performs similarly to a properly tuned SKF controller, outperforming FxNLMS.
- Comprehensive simulation and real-time validation demonstrate that the proposed algorithm outperforms the eIAEKF and adaptive Kalman-like algorithms under gradual dynamic conditions.
- Real-time validation on a hardware platform confirms the computational feasibility, robustness, and practical effectiveness of the proposed algorithm for MIMO ANC in different scenarios.

The remainder of this paper is organized as follows: Section 2 introduces the multichannel ANC model; Section 2.1 reviews the standard Kalman filter; Section 2.3 presents the derivation of the proposed GeIAEKF. Section 3 reports the simulation results, Section 4 reports experimental results, and Section 5 concludes the paper.

Table 1
Frequently used symbols, signals, vectors, and matrices.

Notation	Definition
I	Number of reference sensors.
J	Number of secondary actuators.
K	Number of error sensors.
L	Length (number of taps) of each adaptive control filter.
M	Length (number of taps) of the FIR model of each secondary path.
N	State dimension / total number of adaptive coefficients: $N = LIJ$.
N_w	Sliding-window length used to estimate error covariance.
$x_i(n)$	Reference signal measured at the i-th reference microphone.
$\mathbf{x}(n) \in \mathbb{R}^I$	Reference signal vector.
$\mathbf{x}^L(n) \in \mathbb{R}^{LI}$	Stacked reference signals buffer of length L (cf. (5)).
$y_j(n)$	Control (anti-noise) signal driving the j-th loudspeaker.
$\mathbf{y}(n) \in \mathbb{R}^J$	Control signal vector.
$\mathbf{y}_f(n) \in \mathbb{R}^K$	Anti-noise contribution at the error microphone positions (cf. (8)).
$d_k(n)$	Primary disturbance/ undesired noise measured at the k-th error microphone.
$\mathbf{d}(n) \in \mathbb{R}^K$	Undesired noise vector.
$w_{i,j}(n)$	Adaptive coefficient between $i-th$ reference signal and $j-th$ loudspeaker.
$\mathbf{w}_{Lj}(n) \in \mathbb{R}^{LI}$	Adaptive coefficient vector driving loudspeaker j.
$\mathbf{w}(n) \in \mathbb{R}^{LIJ}$	Stacked adaptive coefficient (state) vector (cf. (4).
$\mathbf{w}^+(n)$	A posteriori estimate of $\mathbf{w}(n)$ (KF estimate).
$\hat{\mathbf{s}}_{jk} \in \mathbb{R}^M$	Estimated secondary-path impulse response from loudspeaker j to error microphone k.
$\mathbf{X}^M(n) \in \mathbb{R}^{I\times M}$	Reference data matrix containing the most recent M samples of each reference signal $x_i(n)$(cf. (6)).
$\hat{\mathbf{x}}_{jk}(n) \in \mathbb{R}^I$	Filtered-x reference vector (cf. (7)).
$\mathbf{H}(n) \in \mathbb{R}^{K\times LIJ}$	Measurement matrix (cf. (9)).
$\mathbf{F}(n) \in \mathbb{R}^{LIJ\times LIJ}$	State transition matrix.
$\mathbf{q}(n) \in \mathbb{R}^{LIJ}$	Process noise vector.
$\mathbf{Q}(n) \in \mathbb{R}^{LIJ\times LIJ}$	Process noise covariance matrix, $\mathbf{Q}(n) = \mathbb{E}\{\mathbf{q}(n)\mathbf{q}^T(n)\}$.
$\mathbf{r}(n) \in \mathbb{R}^K$	Measurement noise vector.
$\mathbf{R}(n) \in \mathbb{R}^{K\times K}$	Measurement noise covariance matrix, $\mathbf{R}(n) = \mathbb{E}\{\mathbf{r}(n)\mathbf{r}^T(n)\}$.
$\mathbf{e}(n) \in \mathbb{R}^K$	Residual error vector measured by the K error microphones.
$\mathbf{P}(n) \in \mathbb{R}^{LIJ\times LIJ}$	State error covariance matrix.
$\hat{\mathbf{C}}_{\mathbf{v}}(n) \in \mathbb{R}^{K\times K}$	Estimated innovation (error) covariance matrix (cf. (23)).

Throughout this paper, bold uppercase letters denote matrices (e.g., $\mathbf{A}$), bold lowercase letters denote column vectors (e.g., $\mathbf{a}$), and italic letters denote scalars (e.g., a). The operator $(\cdot)^T$ denotes transpose. The identity matrix of size A is denoted by $\mathbf{I}_A$, and $\mathbf{0}_{A\times B}$ denotes an $A \times B$ all-zero matrix. The main symbols are summarized in Table 1.

2. MIMO ANC problem formulation

Consider a MIMO ANC system consisting of I reference microphones, J secondary loudspeakers, and K error microphones. The system is modeled in such a way that the reference sensors only get information from the reference signals, ignoring the signals from the secondary loudspeakers, and the control algorithm uses the filtered-x model, as illustrated in Figure 1. This assumption is common in many ANC configurations when the distance between these elements is sufficiently large, as is the case in the systems considered in the experimental part of this work. It should be noted that the objective of this work is to minimize the noise signal at the control points, where the error signals (e_k) are measured. Since these signals require the presence of a microphone, the control points cannot coincide with the actual positions of a user's ears, where control microphones cannot be physically placed. Nevertheless, it can reasonably be assumed that the cancellation effect extends to the vicinity of the control microphones, provided the user is sufficiently close to these positions. If the listening position is not sufficiently close to the control points, remote microphone techniques could alternatively be employed to estimate the noise at the desired listening position (Antoñanzas, Ferrer,

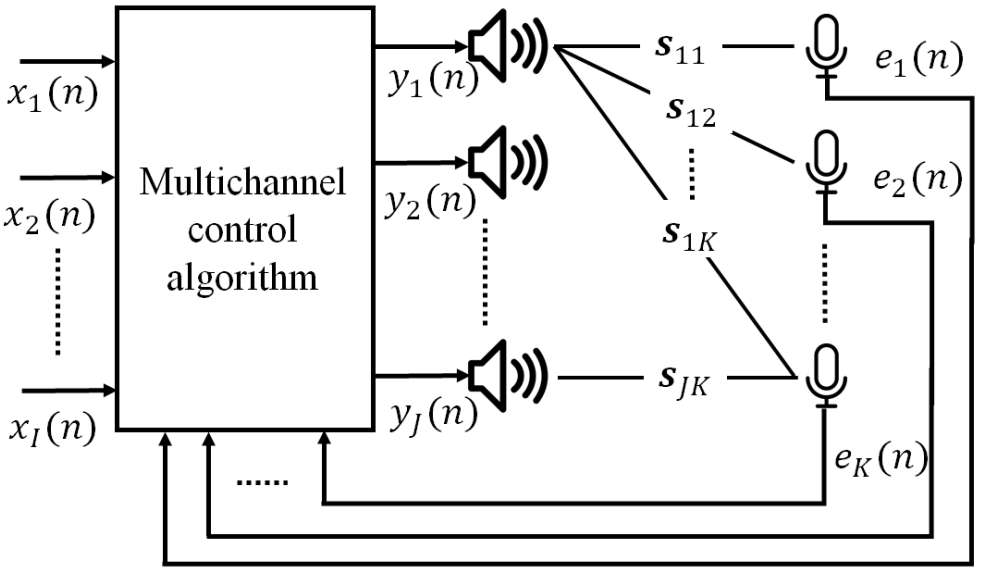


Figure 1: Simplified generic diagram of a MIMO ANC system.

De Diego and Gonzalez, 2023). In this article, $\mathbf{p}(n)$ denotes the primary acoustic paths from the noise source to the error microphones, $\mathbf{s}(n)$ the secondary acoustic paths from the secondary loudspeakers to the error microphones, $\hat{\mathbf{s}}(n)$ the estimate of the secondary paths, and $\mathbf{w}(n)$ the adaptive control filter.

Let us define the reference signal vector $\mathbf{x}(n) \in \mathbb{R}^I$, the control signal vector $\mathbf{y}(n) \in \mathbb{R}^J$, the error signal vector $\mathbf{e}(n) \in \mathbb{R}^K$, the primary disturbance measured at the k-th error microphone, $d_k(n)$ and the undesired noise vector $\mathbf{d}(n) = [d_1(n)\ d_2(n)\ \ldots\ d_K(n)]^T \in \mathbb{R}^K$, where n denotes the discrete-time index. The anti-noise signal at the k-th error microphone generated by J secondary loudspeakers is given by

$$y_{fk}(n) = \sum_{j=1}^{J} [-y_j(n) * s_{jk}(n)], \tag{1}$$

where, $*$ denotes discrete convolution. The stacked anti-noise vector at the error microphones is then

$$\mathbf{y}_f(n) = [y_{f1}(n)\ y_{f2}(n)\ \ldots\ y_{fK}(n)]^T.$$

Here, $s_{jk}(n)$ is the M-length FIR impulse response modeling the secondary path between the j-th loudspeaker and k-th microphone, expressed as a vector: $\mathbf{s}_{jk} = [s_{jk}(0)\ s_{jk}(1)\ \ldots\ s_{jk}(M-1)]^T$. The output vector is $\mathbf{y}(n) = [y_1(n)\ y_2(n)\ \ldots\ y_J(n)]^T$, with each component computed as:

$$y_j(n) = \mathbf{w}_{Lj}^T(n)\mathbf{x}^L(n), \tag{2}$$

where $\mathbf{w}_{Lj}(n) \in \mathbb{R}^{LI}$ contains L adaptive filters that are used to generate the anti-noise signal for the j-th loudspeaker:

$$\begin{aligned} \mathbf{w}_j(n) &= [w_{1,j}(n)\ w_{2,j}(n)\ \ldots w_{I,j}(n)]^T \in \mathbb{R}^I, \\ \mathbf{w}_{Lj}(n) &= [\mathbf{w}_j^T(n)\ \mathbf{w}_j^T(n-1)\ \ldots\ \mathbf{w}_j^T(n-L+1)]^T \in \mathbb{R}^{LI}. \end{aligned} \tag{3}$$

The overall adaptive filter coefficients are stacked as:

$$\mathbf{w}(n) = [\mathbf{w}_{L1}^T(n)\ \mathbf{w}_{L2}^T(n)\ \ldots\ \mathbf{w}_{LJ}^T(n)]^T \in \mathbb{R}^{LIJ}. \tag{4}$$

The reference signal vector $\mathbf{x}^L(n)$ is constructed from the most recent L values of the I reference microphones:

$$\mathbf{x}(n) = [x_1(n)\ x_2(n)\ \ldots\ x_I(n)]^T \in \mathbb{R}^I,$$

$$\mathbf{x}^L(n) \quad = \quad [\mathbf{x}^T(n)\ \mathbf{x}^T(n-1)\ \ldots\ \mathbf{x}^T(n-L+1)]^T \in \mathbb{R}^{LI}. \tag{5}$$

According to the filtered-x scheme of the ANC system, the control algorithm requires the most recent M reference signal vectors $\mathbf{x}(n)$ filtered by each secondary path estimate $\hat{\mathbf{s}}_{jk} \in \mathbb{R}^M$, that is:

$$\mathbf{X}^M(n) = [\mathbf{x}(n)\ \mathbf{x}(n-1) \ldots\ \mathbf{x}(n-M+1)] \in \mathbb{R}^{I\times M}, \quad \text{and} \tag{6}$$

$$\hat{\mathbf{x}}_{jk}(n) = \mathbf{X}^M(n)\hat{\mathbf{s}}_{jk} \ \in \mathbb{R}^I. \tag{7}$$

Assuming the KF algorithm provides an estimate $\mathbf{w}^+(n-1)$ of the overall adaptive filter coefficients $\mathbf{w}(n-1)$, the output control signal vector can be modeled as (Ferrer, de Diego, Piñero and Gonzalez, 2020):

$$\mathbf{y}(n) = -\mathbf{H}(n)\mathbf{w}^+(n-1) \in \mathbb{R}^K, \tag{8}$$

where $\mathbf{H}(n) \in \mathbb{R}^{K\times LIJ}$ is the measurement matrix that is built from the $\hat{\mathbf{x}}_{jk}(n)$ vectors as

$$\mathbf{H}(n) = \begin{bmatrix} \hat{\mathbf{x}}^L_{11}(n) & \hat{\mathbf{x}}^L_{12}(n) & \ldots & \hat{\mathbf{x}}^L_{1K}(n) \\ \hat{\mathbf{x}}^L_{21}(n) & \hat{\mathbf{x}}^L_{22}(n) & \ldots & \hat{\mathbf{x}}^L_{2K}(n) \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\mathbf{x}}^L_{J1}(n) & \hat{\mathbf{x}}^L_{J2}(n) & \ldots & \hat{\mathbf{x}}^L_{JK}(n) \end{bmatrix}^T \tag{9}$$

where $\hat{\mathbf{x}}^L_{jk}(n)$ is a vector that contains the most recent L vectors of $\hat{\mathbf{x}}_{jk}(n)$ as

$$\hat{\mathbf{x}}^L_{jk}(n) = [\hat{\mathbf{x}}^T_{jk}(n)\ \hat{\mathbf{x}}^T_{jk}(n-1)\ \ldots\ \hat{\mathbf{x}}^T_{jk}(n-L+1)]^T \in \mathbb{R}^{LI}.$$

Hence, the MIMO ANC system can be expressed in state-space form as:

$$\mathbf{w}(n) = \mathbf{F}(n-1)\mathbf{w}(n-1) + \mathbf{q}(n-1), \tag{10}$$

$$\mathbf{d}(n) = \mathbf{H}(n)\mathbf{w}(n) + \mathbf{r}(n). \tag{11}$$

where $\mathbf{F}(n-1) \in \mathbb{R}^{LIJ\times LIJ}$ is the state transition matrix, $\mathbf{d}(n)$ is the undesired noise signal, $\mathbf{q}(n-1) \in \mathbb{R}^{LIJ}$ is the process noise vector, and $\mathbf{r}(n) \in \mathbb{R}^K$ is the measurement noise vector.

2.1. Standard Kalman Filter

The Kalman filter estimates $\mathbf{w}(n)$ through prediction and update steps. A priori estimates are denoted with the superscript $^-$ and a posteriori estimates with the superscript $^+$ (Liebich et al., 2017):

Prediction Step:

$$\begin{aligned} \mathbf{w}^-(n) &= \mathbf{F}(n-1)\mathbf{w}^+(n-1), \\ \mathbf{P}^-(n) &= \mathbf{F}(n-1)\mathbf{P}^+(n-1)\mathbf{F}^T(n-1) + \mathbf{Q}(n-1), \end{aligned}$$

Kalman Gain:

$$\begin{aligned} \mathbf{Z}(n) &= \mathbf{P}^-(n)\mathbf{H}^T(n), \\ \mathbf{T}(n) &= \mathbf{H}(n)\mathbf{Z}(n) + \mathbf{R}(n), \\ \mathbf{K}(n) &= \mathbf{Z}(n)\mathbf{T}^{-1}(n), \end{aligned} \tag{12}$$

Update Step:

$$\mathbf{w}^+(n) = \mathbf{w}^-(n) + \mathbf{K}(n)[\mathbf{d}(n) - \mathbf{H}(n)\mathbf{w}^-(n)],$$

$$\mathbf{P}^{+}(n) = \mathbf{P}^{-}(n) - \mathbf{K}(n)\mathbf{H}(n)\mathbf{P}^{-}(n).$$

Where the following definitions are considered: the predicted state error covariance matrix $\mathbf{P}^{-}(n) = \mathbb{E}\{(\mathbf{w}(n) - \mathbf{w}^{-}(n))(\mathbf{w}(n) - \mathbf{w}^{-}(n))^{T}\}$, the updated state error covariance matrix $\mathbf{P}^{+}(n) = \mathbb{E}\{(\mathbf{w}(n) - \mathbf{w}^{+}(n))(\mathbf{w}(n) - \mathbf{w}^{+}(n))^{T}\}$, the measurement noise covariance matrix $\mathbf{R}(n) = \mathbb{E}\{\mathbf{r}(n)\mathbf{r}^{T}(n)\}$, and the process noise covariance matrix $\mathbf{Q}(n) = \mathbb{E}\{\mathbf{q}(n)\mathbf{q}^{T}(n)\}$. These equalities hold only under the assumption that the data in $\mathbf{w}(n)$ is statistically independent of the data in $\mathbf{w}^{-}(n)$ and $\mathbf{w}^{+}(n)$. Moreover, the elements of $\mathbf{H}(n)$ should be deterministic. The prediction uses system dynamics from prior estimates of the state $\mathbf{w}^{+}(n-1)$ and the state error covariance matrix $\mathbf{P}^{+}(n-1)$. The update incorporates the current undesired noise signal $\mathbf{d}(n)$, measurement matrix $\mathbf{H}(n)$ and, predicted filter coefficient $\mathbf{w}^{-}(n)$, adjusted by Kalman gain $\mathbf{K}(n)$, yielding updated state $\mathbf{w}^{+}(n)$ and state error covariance $\mathbf{P}^{+}(n)$.

Algorithm 1: Standard Kalman Filter for MIMO ANC (Naive implementation)

Setup: Set $\hat{\mathbf{s}}_{jk}$, $\mathbf{Q}(0)$, $\mathbf{R}(0)$, $\mathbf{w}^{+}(0)$, and $\mathbf{P}^{+}(0)$.
for *each time step* $n = 1, 2, \ldots,$ **do**
  Input: $x_i(n)$, $e_k(n)$,$\mathbf{w}^{+}(n-1)$, and $\mathbf{P}^{+}(n-1)$.
  1. Reference Expansion and Matrix Construction:
  Form buffer $\mathbf{x}^{M}(n) = [\mathbf{x}(n)\ \mathbf{x}(n-1) \ldots\ \mathbf{x}(n-M+1)]$
  2. Build Measurement Matrix $\mathbf{H}(n)$:
  for $j = 1$ *to* J, $k = 1$ *to* K **do**
    $\hat{\mathbf{x}}_{jk}(n) = \mathbf{x}^{M}(n)\hat{\mathbf{s}}_{jk} \Rightarrow (IM)$ multiplications
    Insert into $\mathbf{H}(n)$
  end
  3. Control Signal Generation:
  for $j = 1$ *to* J **do**
    $y_j(n) = (\mathbf{w}_j^{+}(n-1))^{T}\mathbf{x}^{L}(n) \Rightarrow (N)$
  end
  4. Kalman Prediction:
  $\mathbf{w}^{-}(n) = \mathbf{w}^{+}(n-1)$
  $\mathbf{P}^{-}(n) = \mathbf{P}^{+}(n-1) + \mathbf{Q}(n-1)$
  5. Kalman Gain Computation:
  $\mathbf{Z}(n) = \mathbf{P}^{-}(n)\mathbf{H}^{T}(n) \Rightarrow (N^2K)$
  $\mathbf{T}(n) = \mathbf{H}(n)\mathbf{Z}(n) + \mathbf{R}(n) \Rightarrow (NK^2)$
  $\mathbf{K}(n) = \mathbf{Z}(n)\mathbf{T}^{-1}(n) \Rightarrow (NK^2 + K^3)$
  6. Update:
  $\mathbf{w}^{+}(n) = \mathbf{w}^{-}(n) + \mathbf{K}(n)\mathbf{e}(n) \Rightarrow (NK)$
  $\mathbf{P}^{+}(n) = \mathbf{P}^{-}(n) - \mathbf{K}(n)\mathbf{H}(n)\mathbf{P}^{-}(n) \Rightarrow (N^2K + N^3)$
  Output: $\mathbf{y}(n)$, $\mathbf{w}^{+}(n)$, and $\mathbf{P}^{+}(n)$.
end

In ANC systems, the actual undesired noise signal $\mathbf{d}(n)$ is not directly observable, and only the error signal $\mathbf{e}(n)$ can be measured. The SKF formulation uses $\mathbf{d}(n)$, which would require either modifying the system model or explicitly estimating $\mathbf{d}(n)$, both of which increase computational complexity and may compromise real-time operation. Thus, the direct use of the $\mathbf{e}(n)$ makes the algorithm more suitable for the ANC applications. For this reason, we approximate $[\mathbf{d}(n) - \mathbf{H}(n)\mathbf{w}^{+}(n-1)] \approx \mathbf{e}(n)$ and assume $\mathbf{F}(n) = \mathbf{I}_{LIJ}$ so that the prediction simplifies to:

$$\mathbf{w}^{-}(n) = \mathbf{w}^{+}(n-1),$$
$$\mathbf{P}^{-}(n) = \mathbf{P}^{+}(n-1) + \mathbf{Q}(n-1).$$

The pseudocode of the SKF algorithm for MIMO ANC, along with the number of multiplications per operation as a measure of computational burden, is provided in Algorithm 1.

Typically, $\mathbf{F}(n)$, $\mathbf{Q}(n)$, and $\mathbf{R}(n)$ are kept fixed and tuned manually. One of the biggest challenges of the SKF is to tune the $\mathbf{Q}(n)$, and $\mathbf{R}(n)$ matrices. Especially in an acoustic system, it is quite hard to predict or know the process and

measurement noise throughout the process. Additionally, the process noise and measurement noise can change over time due to changes in the process and system dynamics. Hence, the KF algorithm with adaptive covariance parameters is more suitable for acoustic applications due to the acoustic system's behavior and characteristics. However, adapting covariance parameters increases the computational burden of the already computationally complex KF algorithm. In the next sections, some computationally efficient adaptive Kalman filters for MIMO ANC systems are presented.

2.2. The eIAEKF Algorithm for MIMO ANC

The IAEKF algorithm enhances the estimation of the covariance matrices $\mathbf{Q}(n)$ and/or $\mathbf{R}(n)$ under uncertain or suboptimal filtering conditions. For ANC systems operating in dynamic acoustic environments, adaptively updating these matrices is essential for maintaining robust real-time performance.

The IAEKF utilizes the innovation sequence, defined as the difference between the actual desired signal and its predicted value:

$$\mathbf{v}(n) = \mathbf{d}(n) - \widehat{\mathbf{d}}(n), \tag{13}$$

where the predicted desired signal can be written by projecting the prior state estimate:

$$\widehat{\mathbf{d}}(n) = \mathbf{H}(n)\mathbf{w}^{-}(n), \tag{14}$$

Substituting (14) into (13) yields:

$$\mathbf{v}(n) = \mathbf{d}(n) - \mathbf{H}(n)\mathbf{w}^{-}(n), \tag{15}$$

The innovation sequence $\mathbf{v}(n)$ captures new information beyond what is predicted by the prior state and is temporally uncorrelated (Kailath, 1972). From the system model (11):

$$\mathbf{v}(n) \approx \mathbf{H}(n)(\mathbf{w}(n) - \mathbf{w}^{-}(n)) + \mathbf{r}(n). \tag{16}$$

The covariance of $\mathbf{v}(n)$ is:

$$\mathbf{C}_{\mathbf{v}}(n) = \mathbb{E}\{\mathbf{v}(n)\mathbf{v}^{T}(n)\} = \mathbf{H}(n)\mathbf{P}^{-}(n)\mathbf{H}^{T}(n) + \mathbf{R}(n) = \mathbf{T}(n). \tag{17}$$

Rearranged, this allows estimation of $\mathbf{R}(n)$:

$$\mathbf{R}(n) = \mathbf{C}_{\mathbf{v}}(n) - \mathbf{H}(n)\mathbf{P}^{-}(n)\mathbf{H}^{T}(n). \tag{18}$$

In practice, $\mathbf{C}_{\mathbf{v}}(n)$ can be approximated via a moving average or sliding window of size $N_w = LIJ$ (Xu, Su and Liu, 2014):

$$\hat{\mathbf{C}}_{\mathbf{v}}(n) = \frac{1}{N_w} \sum_{m=n-N_w+1}^{n} \mathbf{v}(m)\mathbf{v}^{T}(m). \tag{19}$$

The residual error $\mathbf{e}(n)$ generated by the interference between the undesired noise $\mathbf{d}(n)$ and the anti-noise $\mathbf{y}_f(n)$, can be expressed using (11) and (8) as:

$$\begin{aligned} \mathbf{e}(n) &\approx \mathbf{d}(n) + \mathbf{y}_f(n), \\ &\approx \mathbf{H}(n)(\mathbf{w}(n) - \mathbf{w}^{+}(n-1)) + \mathbf{r}(n), \end{aligned} \tag{20}$$

where $\mathbf{w}^{+}(n-1)$ are the adaptive filter coefficients estimated by the control algorithm and $\mathbf{w}(n)$ are the adaptive filter coefficients defined in (10).

Consequently, the innovation sequence is derived from (16) and (20) as follows:

$$\mathbf{v}(n) \approx \mathbf{e}(n). \tag{21}$$

Therefore, the innovation covariance matrix can be estimated from the error signals by

$$\hat{\mathbf{C}}_{\mathbf{v}}(n) \approx \frac{1}{N_w} \sum_{m=n-N_w+1}^{n} \mathbf{e}(m)\mathbf{e}^{T}(m), \tag{22}$$

where adding a regularization term helps improve numerical stability:

$$\hat{\mathbf{C}}_{\mathbf{v}}(n) \approx \left(\frac{1}{N_w} \sum_{m=n-N_w+1}^{n} \mathbf{e}(m)\mathbf{e}^T(m) \right) + \epsilon \mathbf{I}_K, \tag{23}$$

where ϵ is a small constant.
This time-varying innovation covariance is then used to estimate the Kalman gain by

$$\mathbf{K}(n) = \mathbf{P}^-(n)\mathbf{H}^T(n)\hat{\mathbf{C}}_{\mathbf{v}}^{-1}(n) = \mathbf{Z}(n)\hat{\mathbf{C}}_{\mathbf{v}}^{-1}(n). \tag{24}$$

The regularized estimate $\hat{\mathbf{C}}_{\mathbf{v}}(n)$ is symmetric positive definite in practice, allowing its inverse to be efficiently computed via Cholesky factorization rather than direct matrix inversion. This algorithm was originally proposed and validated in our previous work (Islam, Ferrer, de Diego and Gonzalez, 2026). This algorithm exhibits a lower computational burden and superior performance compared to the available adaptive Kalman-like algorithm for ANC applications. However, this algorithm requires manual tuning of the $\mathbf{Q}(n)$, and the computational complexity is still quite high. So, we propose and derive the extended generalized version of the eIAEKF algorithm with $\mathbf{Q}(n)$ adaptation and lower computational cost.

2.3. Generalized eIAEKF Algorithm for MIMO ANC

The most computationally demanding operation of the KF is updating the state error covariance matrix, $\mathbf{P}^+(n) = \mathbf{P}^-(n) - \mathbf{K}(n)\mathbf{H}(n)\mathbf{P}^-(n)$.

The state error covariance matrix $\mathbf{P}(n) = \mathbb{E}\{\varepsilon(n)\varepsilon^T(n)\} \in \mathbb{R}^{N\times N}$ represents the correlation matrix of the estimated error associated with the filtered $\mathbf{w}^+(n)$.

The a posteriori filtered state-error vector $\varepsilon^+(n)$ is defined as the difference between the adaptive filter coefficients $\mathbf{w}(n)$ and their estimated values $\mathbf{w}^+(n)$.

$$\varepsilon^+(n) = \mathbf{w}(n) - \mathbf{w}^+(n) = \mathbf{w}(n) - \mathbf{w}^-(n) - \mathbf{K}(n)\mathbf{e}(n) = \varepsilon^-(n) - \mathbf{K}(n)\mathbf{e}(n). \tag{25}$$

where $\varepsilon^-(n)$ is the predicted state error vector at time n.
So, from eq. (25) we can write:

$$\mathbf{P}^+(n) = \mathbb{E}\{\varepsilon^+(n)(\varepsilon^+(n))^T\}. \tag{26}$$

Note that $\mathbf{K}(n)$ is treated as deterministic with respect to the expectation operator $\mathbb{E}\{.\}$, since it depends on $\mathbf{P}^-(n)$, $\mathbf{H}^T(n)$, and $\hat{\mathbf{C}}_{\mathbf{v}}^{-1}(n)$ that are assumed to be known. Consequently, the only random quantity in the eq. (25) is $\mathbf{e}(n)$ defined in eq. (20) as both $\mathbf{r}(n)$ and $\mathbf{w}(n)$ are random processes. A similar conclusion can be found in (Haykin, 2002; Anderson and Moore, 2005). Therefore, the eq. (26) can be approximated as:

$$\mathbf{P}^+(n) \approx \mathbb{E}\{\varepsilon^-(n)(\varepsilon^-(n))^T\} + \mathbf{K}(n)\mathbb{E}\{\mathbf{e}(n)\mathbf{e}^T(n)\}\mathbf{K}^T(n) - 2\mathbb{E}\{\varepsilon^-(n)\mathbf{e}^T(n)\}\mathbf{K}^T(n). \tag{27}$$

Examining the right-hand side of (27), it contains three expectations, which can be interpreted individually as follows:

- The first expectation corresponds to the predicted state-error covariance matrix:

$$\mathbf{P}^-(n) = \mathbb{E}\{\varepsilon^-(n)(\varepsilon^-(n))^T\}.$$

- The expectation in the second term represents the correlation matrix of the residual error $\mathbf{e}(n)$:

$$\hat{\mathbf{C}}_{\mathbf{v}}(n) \approx \mathbb{E}\{\mathbf{e}(n)\mathbf{e}^T(n)\}.$$

- The expectation in the third term may be rewritten as:

$$\mathbb{E}\{\varepsilon^-(n)\mathbf{e}^T(n)\} = \mathbb{E}\{(\mathbf{w}(n) - \mathbf{w}^-(n))\mathbf{e}^T(n)\} = \mathbb{E}\{\mathbf{w}(n)\mathbf{e}^T(n)\},$$

where the last equality follows from the orthogonality assumption between $\mathbf{w}^-(n) = \mathbf{w}^+(n-1)$ and the residual error vector $\mathbf{e}(n)$, which serves as the input.

The update process of the Kalman filter depends on prior knowledge (Haykin, 2002). Accordingly, the a posteriori estimate of $\mathbf{w}^+(n)$ can be expressed as

$$\begin{aligned}\mathbf{w}^+(n) &= \sum_{k=1}^{n-1}(\mathbb{E}\{\mathbf{w}(n)\mathbf{e}^T(k)\}\mathbf{C}_{\mathbf{v}}^{-1}(k)\mathbf{e}(k)) + \mathbb{E}\{\mathbf{w}(n)\mathbf{e}^T(n)\}\mathbf{C}_{\mathbf{v}}^{-1}(n)\mathbf{e}(n) \\ &= \mathbf{w}^-(n) + \mathbf{K}(n)\mathbf{e}(n), \end{aligned} \tag{28}$$

where, $\mathbf{K}(n) = \mathbb{E}\{\mathbf{w}(n)\mathbf{e}^T(n)\}\mathbf{C}_{\mathbf{v}}^{-1}(n)$. Consequently, we can assume $\mathbb{E}\{\mathbf{w}(n)\mathbf{e}^T(n)\} \approx \mathbf{K}(n)\mathbf{C}_{\mathbf{v}}(n)$.
Substituting this result into (27), the a posteriori state-error covariance matrix can be expressed as:

$$\begin{aligned}\mathbf{P}^+(n) &= \mathbf{P}^-(n) + \mathbf{K}(n)\mathbf{C}_{\mathbf{v}}(n)\mathbf{K}^T(n) - 2\mathbf{K}(n)\mathbf{C}_{\mathbf{v}}(n)\mathbf{K}^T(n) \\ &= \mathbf{P}^-(n) - \mathbf{K}(n)\mathbf{C}_{\mathbf{v}}(n)\mathbf{K}^T(n) \\ &= \mathbf{P}^-(n) - \mathbf{Z}(n)\mathbf{K}^T(n). \end{aligned} \tag{29}$$

For the generalized eIAEKF (GeIAEKF) approach, the Kalman gain is calculated as $\mathbf{K}(n) = \mathbf{Z}(n)\mathbf{C}_{\mathbf{v}}^{-1}(n)$ from which we can write $\mathbf{K}(n)\mathbf{C}_{\mathbf{v}}(n) = \mathbf{Z}(n)$.

In the GeIAEKF Algorithm, the process noise covariance matrix is adapted following the strategy used in the adaptive Kalman-like algorithm reported in (Liebich et al., 2017). Initially, this adaptive step was proposed for the active echo cancellation (AEC) system (Paleologu, Benesty, Ciochină and Grant, 2014), where $\mathbf{Q}(n)$ is adapted based on the deviation between the a priori and a posteriori estimates of $\mathbf{w}(n)$, as the process noise resides in the state deviation. Later, in (Liebich et al., 2017), this insight from the field of AEC is directly transferred to the ANC field, and the term $[\mathbf{d}(n) - \mathbf{H}(n)\mathbf{w}^-(n)]$ is replaced by the measured error signal $\mathbf{e}(n)$ while adapting the transition matrix $\mathbf{F}(n)$. In ANC system, $\mathbf{e}(n)$ is generated by the interaction between the undesired noise $\mathbf{d}(n)$ and the anti-noise $\mathbf{y}_f(n) = -\mathbf{H}(n)\mathbf{w}^+(n-1)$. Therefore, when $\mathbf{F}(n)$ differs from the identity matrix, $\mathbf{w}^-(n) \neq \mathbf{w}^+(n-1)$, and the approximation $[\mathbf{d}(n) - \mathbf{H}(n)\mathbf{w}^-(n)] \approx \mathbf{e}(n)$ no longer strictly holds. In this work, we used this adaptation by considering $\mathbf{F}(n) = I$ to make it more suitable for ANC systems and compatible with error innovation adaptation. Thus the approximation $\mathbf{w}^+(n) - \mathbf{w}^-(n) \approx \mathbf{w}^+(n) - \mathbf{w}^+(n-1) \approx \mathbf{K}(n)\mathbf{e}(n)$. As a result, the $\mathbf{Q}(n)$ is updated using a low-pass IIR filter with smoothing parameter $0 \ll \alpha \leq 1$:

$$\begin{aligned}\mathbf{Q}(n) &= \alpha\mathbf{Q}(n-1) + (1-\alpha)\ diag[\mathbf{w}^+(n) - \mathbf{w}^-(n)][\mathbf{w}^+(n) - \mathbf{w}^-(n)]^T \\ &\approx \alpha\mathbf{Q}(n-1) + (1-\alpha)\ diag[\mathbf{w}^+(n) - \mathbf{w}^+(n-1)][\mathbf{w}^+(n) - \mathbf{w}^+(n-1)]^T \\ &\approx \alpha\mathbf{Q}(n-1) + (1-\alpha)\ diag[\mathbf{K}(n)\mathbf{e}(n)][\mathbf{K}(n)\mathbf{e}(n)]^T. \end{aligned} \tag{30}$$

Equation (30) shows that $\mathbf{Q}(n)$ is adapted solely based on the current measured error signal. This behavior typically yields large covariance values at the initial stages of the control process, resulting in fast convergence and good tracking performance. However, once the algorithm reaches its steady state and the mean of the diagonal elements of $\mathbf{Q}(n)$ becomes small, sudden changes in the system dynamics may lead to a degrading tracking response of the control algorithm. To alleviate this issue, a threshold covariance matrix $\mathbf{Q}_{th}$ is introduced. Whenever the adapted $\mathbf{Q}(n)$ falls below this threshold, it is set to $\mathbf{Q}_{th}$; otherwise, the algorithm will use the adapted $\mathbf{Q}(n)$ value. The value of the threshold covariance matrix,$\mathbf{Q}_{th}$, can be selected based on the system's acoustic path and the acceptable level of residual noise. If we opt for a higher value of $\mathbf{Q}_{th}$, tracking will be faster, but the attenuation level will be lower. In this article, we have chosen 0.01% of the mean energy of the estimated acoustic path as our threshold.

The pseudocode in Algorithm 2 summarizes the proposed algorithm and provides the number of multiplications required per operation.

Algorithm 2: Generalized eIAEKF for MIMO ANC

Setup: Set $\hat{\mathbf{s}}_{jk}$, $\mathbf{w}^{+}(0)$, $\mathbf{P}^{+}(0)$, $\mathbf{Q}_{th}$, $\mathbf{Q}(0)$, ϵ, α, and N_w.

for *each time step* $n = 1, 2, \ldots,$ **do**

Input: $x_i(n)$, $e_k(n)$, $\mathbf{w}^{+}(n-1)$, $\mathbf{P}^{+}(n-1)$, $\hat{\mathbf{C}}_{\mathbf{v}}(n-1)$, and $\mathbf{Q}(n-1)$.

1. Reference Expansion and Matrix Construction:

Form buffer $\mathbf{x}^{M}(n) = [\mathbf{x}(n), \ldots, \mathbf{x}(n-M+1)]$

2. Build Measurement Matrix $\mathbf{H}(n)$:

for $j = 1$ *to* J, $k = 1$ *to* K **do**

$\hat{\mathbf{x}}_{jk}(n) = \mathbf{x}^{M}(n)\hat{\mathbf{s}}_{jk} \Rightarrow (IM)$

Insert into $\mathbf{H}(n)$

end

3. Control Signal Generation:

for $j = 1$ *to* J **do**

$y_j(n) = (\mathbf{w}_j^{+}(n-1))^{T}\mathbf{x}^{L}(n) \Rightarrow (N)$

end

4. Kalman Prediction:

$\mathbf{w}^{-}(n) = \mathbf{w}^{+}(n-1)$

$\mathbf{P}^{-}(n) = \mathbf{P}^{+}(n-1) + \mathbf{Q}(n-1)$

5. Estimate Innovation Covariance:

Computed using sliding window method:

$$\hat{\mathbf{C}}_{\mathbf{v}}(n) \approx \frac{1}{N_w} \sum_{m=n-N_w+1}^{n} \mathbf{e}(m)\mathbf{e}^{T}(m) + \epsilon\mathbf{I}_K \;\Rightarrow (2K^2 + K)$$

6. Kalman Gain Computation:

$\mathbf{Z}(n) = \mathbf{P}^{-}(n)\mathbf{H}^{T}(n) \;\Rightarrow (N^2K)$

$\mathbf{K}(n) = \mathbf{Z}(n)\hat{\mathbf{C}}_{\mathbf{v}}^{-1}(n) \;\Rightarrow (NK^2 + K^2)$

7. Update:

$\mathbf{u}(n) = \mathbf{K}(n)\mathbf{e}(n) \;\Rightarrow (NK)$

$\mathbf{w}^{+}(n) = \mathbf{w}^{-}(n) + \mathbf{u}(n)$

$\mathbf{P}^{+}(n) = \mathbf{P}^{-}(n) - \mathbf{Z}(n)\mathbf{K}^{T}(n) \;\Rightarrow (N^2K)$

8. Adaptive Noise Covariance Updates:

$\mathbf{Q}(n) = \alpha\mathbf{Q}(n-1) + (1-\alpha)\, diag[\mathbf{u}(n)\mathbf{u}^{T}(n)] \Rightarrow (N^2 + 2N)$

if $mean\{diag\{\mathbf{Q}(n)\}\} < mean\{diag\{\mathbf{Q}_{th}\}\}$ **then**

$\mathbf{Q}(n) \leftarrow \mathbf{Q}_{th}$

end

Output: $\mathbf{y}(n)$, $\mathbf{w}^{+}(n)$, $\mathbf{P}^{+}(n)$, $\hat{\mathbf{C}}_{\mathbf{v}}(n)$, and $\mathbf{Q}(n)$.

end

3. Simulation Results

This section presents numerical simulations that evaluate the proposed GeIAEKF algorithm in comparison with the normalized LMS (NLMS) Kim, Hur and Park (2020); Buttarazzi et al. (2024) and the SKF for multichannel ANC systems. Let $N = LIJ$ denote the total number of adaptive filter coefficients. We consider a representative configuration with $L = 200$, $I = 1$, $J = 2$, $K = 2$, and $M = 100$, which yields $N = LIJ = 400$. Unless otherwise stated, this configuration is used.

The secondary paths were identified using standard system identification techniques and remained fixed throughout the adaptation process. The primary disturbance was generated as band-pass-filtered white noise in the 100–900 Hz range to emulate realistic acoustic disturbances. The sampling frequency was set to $f_s = 2000$ Hz for all simulations.

The NLMS algorithm uses an initial step size $\mu = 0.1$ Initialization parameters for the SKF control are $\mathbf{Q}(n) = (1 \times 10^{-8})\mathbf{I}_N$, $\mathbf{R}(n) = (25 \times 10^{-6})\mathbf{I}_K$, $\mathbf{P}(0) = (5 \times 10^{-5})\mathbf{I}_N$, and $\mathbf{w}(0) = \mathbf{0}$. The GeIAEKF control does not require initialization of the process and measurement noise covariance; we set the initial values $\mathbf{Q}(0) = \mathbf{0}$, $\mathbf{R}(0) = \mathbf{I}_K$, $\mathbf{P}(0) = (5 \times 10^{-5})\mathbf{I}_N$, $\mathbf{Q}_{th} = (1 \times 10^{-8})\mathbf{I}_N$, $\alpha = 0.9999$, and $\epsilon = 0.0000001$.

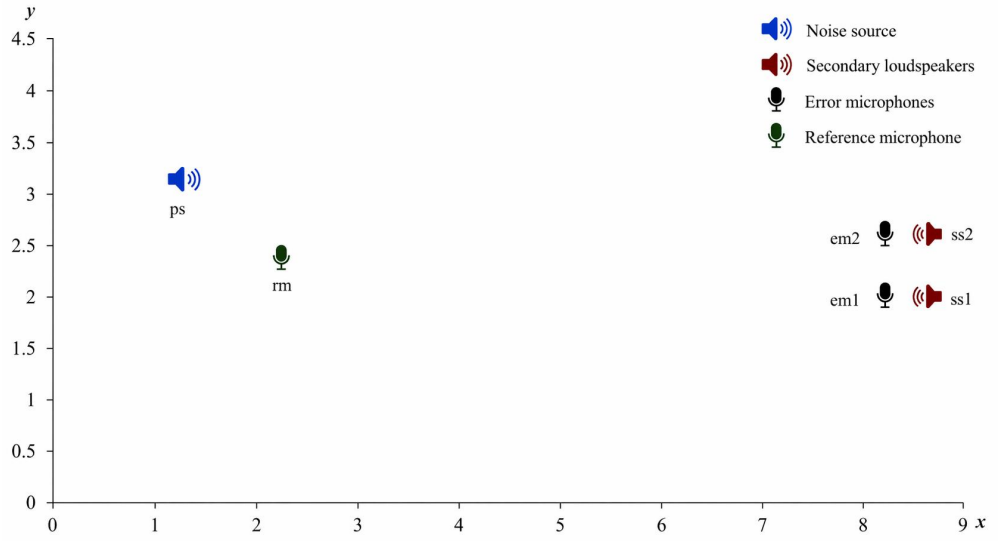


Figure 2: MIMO setup in a room of dimensions $9.13 \times 4.48 \times 2.64$ m^3. Device locations: ps = [1.2, 3.2, 1.5] m, rm = [2.2, 2.3, 1.5] m, $ss1$ = [8.5, 1.94, 1.5] m, $ss2$ = [8.5, 2.54, 1.5] m, $em1$ = [8.3, 2.04, 1.5] m, $em2$ = [8.3, 2.44, 1.5] m.

Performance was evaluated in terms of convergence speed and attenuation, which is calculated as follows,

$$A(n) = 10 \log_{10} \left(\frac{\mathbb{E}\{e^2(n)\}}{\mathbb{E}\{d^2(n)\}} \right), \tag{31}$$

where $d(n)$ denotes the undesired noise and $e(n)$ represents the residual error. The expectation operator $\mathbb{E}\{\cdot\}$ is approximated by time averaging over each simulation run. Each experiment was repeated 100 times with different random seeds, and the attenuation curves were ensemble-averaged to obtain smooth and statistically representative results.

3.1. MIMO Setup for Static Condition

Figure 2 illustrates the MIMO simulation environment. The room dimensions were $9.13 \times 4.48 \times 2.64$ m^3 with a reverberation time of 0.2 s. A multichannel 1x2x2 setup was considered, with one primary noise source (ps), one reference microphone (rm), two secondary loudspeakers ($ss1$ and $ss2$), and two error microphones ($em1$ and $em2$. Acoustic feedback from the secondary sources to the reference microphone was neglected.

The objective was to minimize the residual error signals at both error microphones through adaptation of $\mathbf{w}(n)$. All simulations were conducted using MATLAB in the time domain.

3.1.1. Response to Stationary Primary Noise

The performance of the NLMS, SKF, and the proposed GeIAEKF algorithms was first evaluated under stationary filtered white-noise excitation. The time-domain signals in Fig. 3 show that all three controllers successfully attenuate the disturbance; however, clear differences are evident in their convergence rates and steady-state residual levels.

From the attenuation curves shown in Fig. 4, the SKF achieves the highest steady-state attenuation, reaching approximately -26 dB at $em1$ and -27 dB at $em2$. The GeIAEKF closely follows this performance, stabilizing at approximately -25 to -26 dB, whereas the NLMS algorithm is limited to about -20 dB at the same time.

Fig. 5 illustrates the frequency domain analysis of the residual signals. It confirms the time-domain considerations. The SKF achieves the lowest residual PSD across the 100–900 Hz band, while the GeIAEKF closely follows, typically within $\sim$ 2–4 dB. NLMS exhibits significantly higher residual energy across the entire spectrum.

Overall, under stationary conditions, the SKF provides the best absolute attenuation. However, the GeIAEKF achieves comparable performance (within $\sim$ 5–8 dB), while offering fast convergence and eliminating the need for manual covariance tuning.

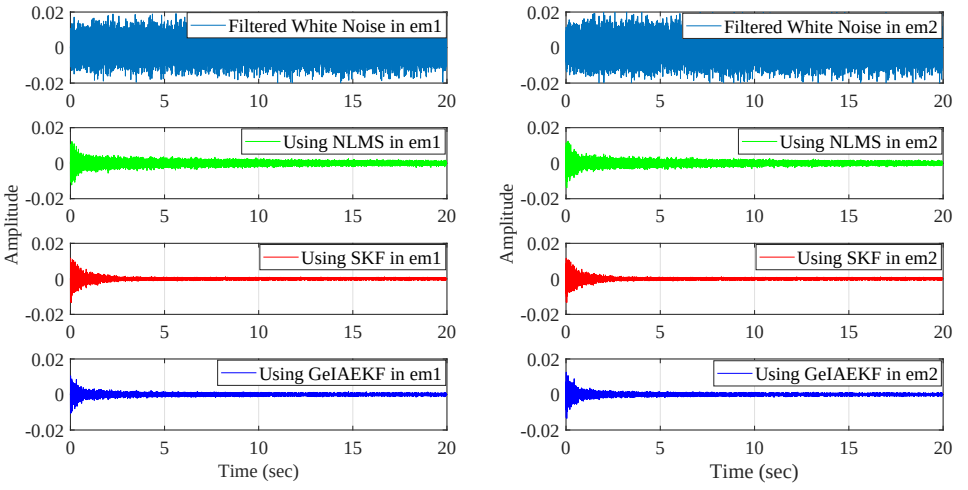


Figure 3: Primary noise and error signals of the MIMO ANC system for stationary noise.

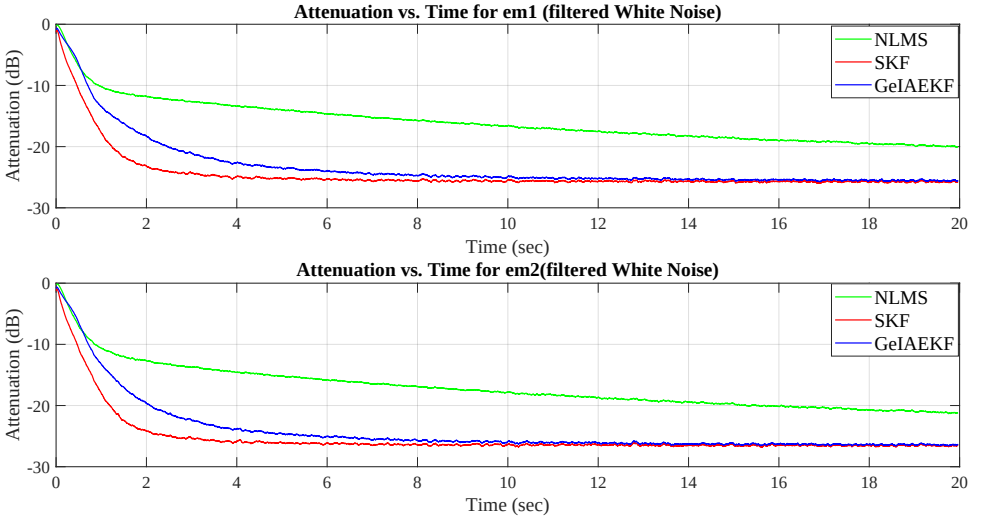


Figure 4: Attenuation of the control algorithms of the MIMO ANC system for stationary noise.

3.2. SISO Setup for Dynamic Conditions

This section presents simulation results for different noise types and noise-source locations. For simplicity, a SISO setup is considered, and the room conditions were consistent with those described in subsection 3.1, focusing solely on the secondary loudspeaker $ss1$ and the error microphone $em1$. The objective was to minimize the error signal $\mathbf{e}_1(n)$ by appropriately adapting the filter coefficients $\mathbf{w}(n)$. The noise source location was varied along the x- and y-axes.

3.2.1. Response to Different Types of Primary Noise

To evaluate robustness under non-stationary conditions, the controller was tested using a composite signal consisting of six consecutive 1.5-second segments (total 9 seconds):

1. Filtered white noise.
2. Colored noise (pink).
3. Periodic tonal noise (sum of 3 sinusoids of 150 Hz, 320 Hz, and 610 Hz).

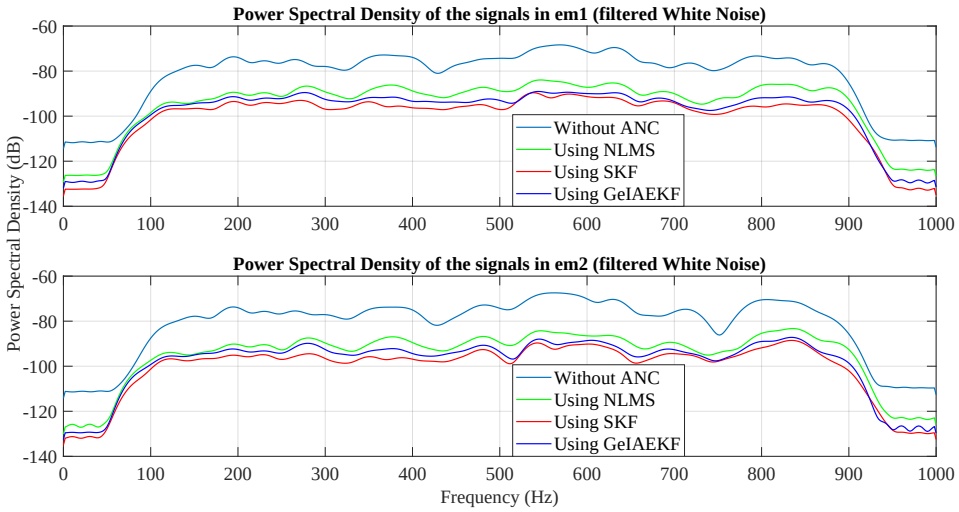


Figure 5: Power spectral density (PSD) of error signals of the MIMO ANC system for stationary noise.

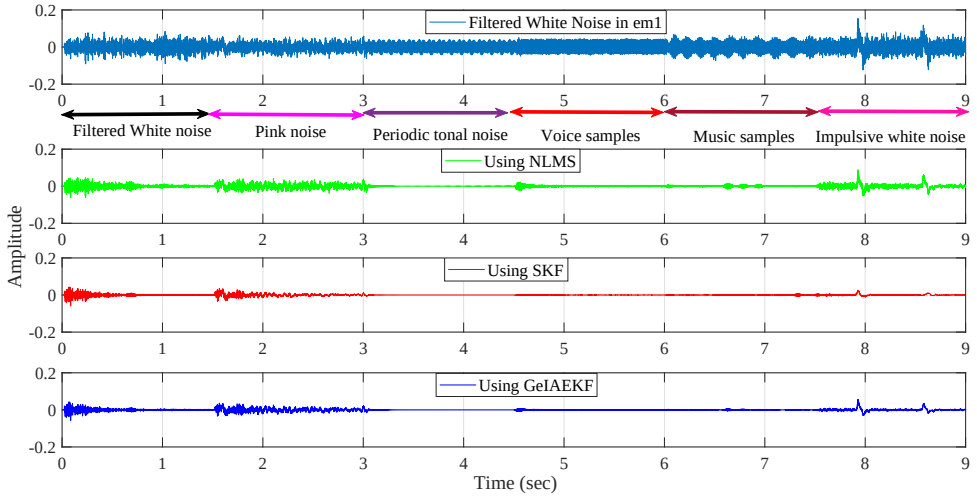


Figure 6: Primary noise and error signals of the MIMO ANC system for different types of noise.

4. Voice sample (synthetic vowel-like voice).
5. Music sample (simple melody with chordal synthesis).
6. Impulsive white noise containing two 100-sample impulses with amplitude five times larger than the background noise.

The composite signal introduces abrupt spectral transitions at each segment boundary, which challenge the adaptation and the controllers' tracking capability. The time-domain responses in Fig. 6 show that NLMS exhibits significant residual energy, particularly during colored noise, speech, and music segments, and large spikes during impulsive disturbances.

The properly tuned SKF shows very good noise suppression and faster tracking capability. However, the GeIAEKF demonstrates almost the same performance as the SKF without requiring prior knowledge. During impulsive disturbances, the GeIAEKF reduces peak residual amplitudes compared to NLMS and achieves performance comparable to that of the SKF.

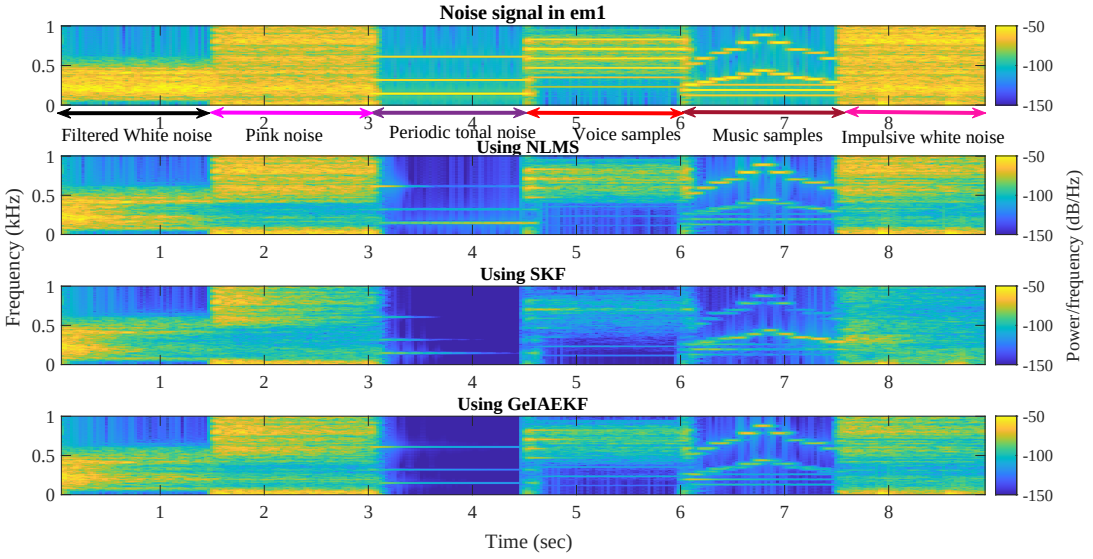


Figure 7: Spectrogram of the residual noise obtained at the *em*1 sensor using the NLMS, SKF, and the GeIAEKF algorithms in the MIMO ANC system, canceling different types of noise.

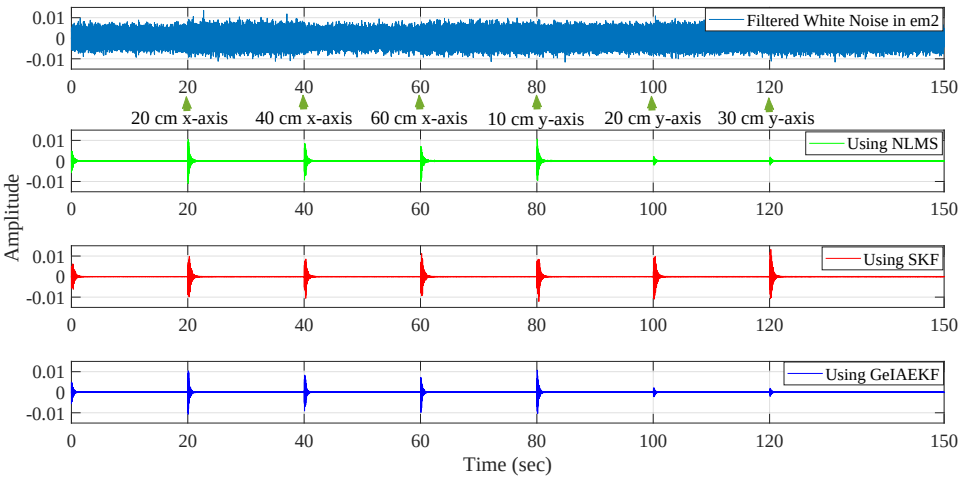


Figure 8: Primary noise and error signals measured at the error microphone (*em*1) using the NLMS, SKF, and GeIAEKF algorithms while changing the noise source (*ps*1) location by 20 cm after 20 seconds, 40 cm after 40 seconds, and 60 cm after 60 seconds along the x-axis and 10 cm after 80 seconds, 20 cm after 100 seconds, and 30 cm after 120 seconds along the y-axis in the SISO ANC system.

The spectrogram analysis in Fig. 7 shows that SKF achieves the strongest suppression of tonal components, while the GeIAEKF closely follows with comparable attenuation. In contrast, the NLMS exhibits persistent spectral leakage.

Overall, although the SKF provides slightly better steady-state suppression in certain segments, the GeIAEKF offers a superior trade-off between tracking speed and attenuation, without requiring parameter tuning.

3.2.2. Response to Noise Source Relocation

To evaluate tracking capability under structural changes, the primary noise source was displaced incrementally along the x- and y-axes, as illustrated in Fig. 8. The source was moved by 20 cm, 40 cm, and 60 cm along the x-axis,

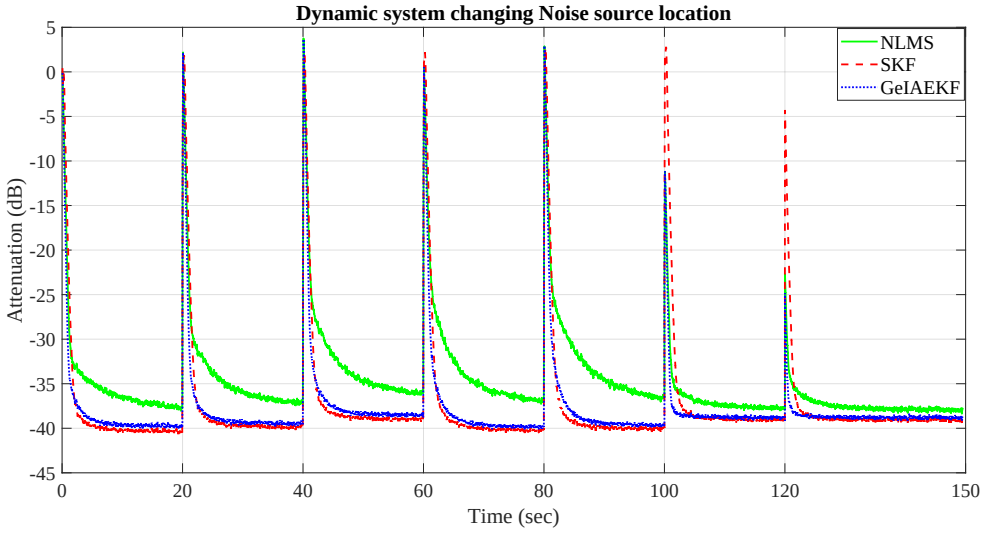


Figure 9: Attenuation using the NLMS, SKF, and GeIAEKF algorithms while changing the noise source ($ps1$) location by 20 cm after 20 seconds, 40 cm after 40 seconds, and 60 cm after 60 seconds along the x-axis and 10 cm after 80 seconds, 20 cm after 100 seconds, and 30 cm after 120 seconds along the y-axis in the SISO ANC system.

followed by 10 cm, 20 cm, and 30 cm along the y-axis. Each relocation introduces an abrupt modification of the primary acoustic path, requiring rapid re-adaptation of the control filter.

The time-domain responses in Fig. 8 show that all controllers experience transient spikes at each relocation instant. However, apparent differences are observed in both recovery speed and transient amplitude. The NLMS exhibits the largest residual peaks and the longest recovery intervals following each displacement. The SKF reduces the transient duration but still shows pronounced overshoot immediately after each change. Additionally, SKF requires that the measured error signals be followed and that all parameters be reinitialized when any sudden changes occur. This operation introduces an additional computational burden.

In contrast, the GeIAEKF demonstrates faster recovery and reduced transient amplitude. After each relocation, the residual error settles to its steady-state level within approximately 0.1×10^5 samples. In contrast, the NLMS requires nearly twice that duration to re-stabilize. The SKF exhibits comparable convergence speed but shows larger transient oscillations immediately after relocation, indicating sensitivity to fixed covariance parameters.

The attenuation curves in Fig. 9 further confirm these observations. Following each relocation event, the proposed method re-establishes attenuation more rapidly and with less overshoot than the competing algorithms.

Overall, these results demonstrate that the GeIAEKF provides robust tracking even under abrupt variations in the primary path. By adapting covariance parameters online using only the measured error signal, the controller maintains high attenuation without requiring manual re-tuning, which is critical for practical ANC applications involving moving noise sources or dynamic environments.

3.2.3. Response to Gradual System Dynamics

In practical ANC systems, variations in the acoustic paths and in the spectral content of the disturbance mostly occur gradually and continuously. To assess the incremental benefit of the generalized error innovation-based adaptive estimation Kalman filter (GeIAEKF), we compare it with the previously proposed eIAEKF Islam et al. (2026) and the adaptive Kalman-like approach Liebich et al. (2017) in two scenarios: a gradually varying primary acoustic path (Fig. 10) and non-stationary road noise (Fig. 11). Unlike comparisons against a fixed-covariance standard Kalman filter, these tests directly examine the added value of process-covariance adaptation relative to adaptive Kalman-based alternatives.

In the case of a gradually varying primary acoustic path, which might happen due to temperature variation, control point, or primary noise displacement, Fig 10 shows that GeIAEKF reaches -25 dB in 2.595 s, compared with 3.779 s for eIAEKF and 3.604 s for the adaptive Kalman-like algorithm. During the final 5 s, GeIAEKF maintains an average attenuation of -26.99 dB, compared with -26.82 dB and -25.08 dB, respectively, indicating faster initial convergence

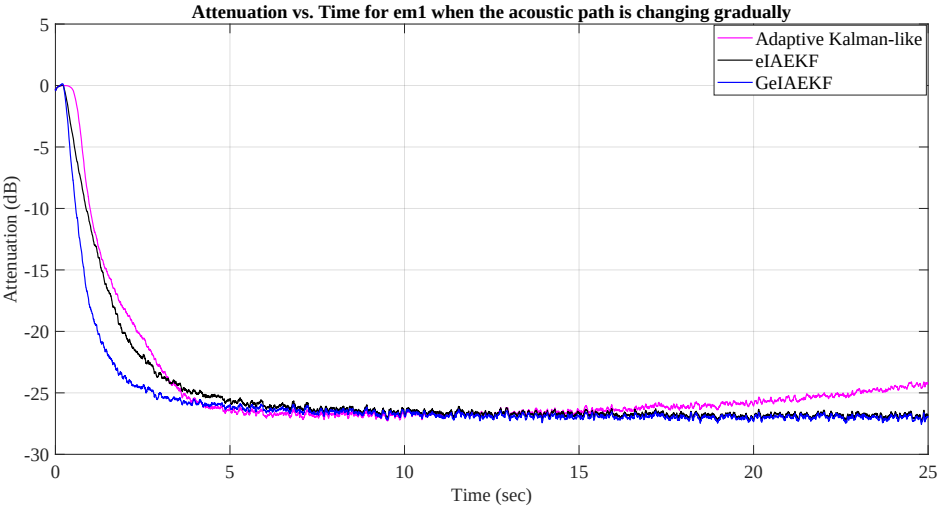


Figure 10: Attenuation using the Adaptive Kalman-like, eIAEKF, and GeIAEKF algorithms while gradually changing the primary acoustic path in the SISO ANC system.

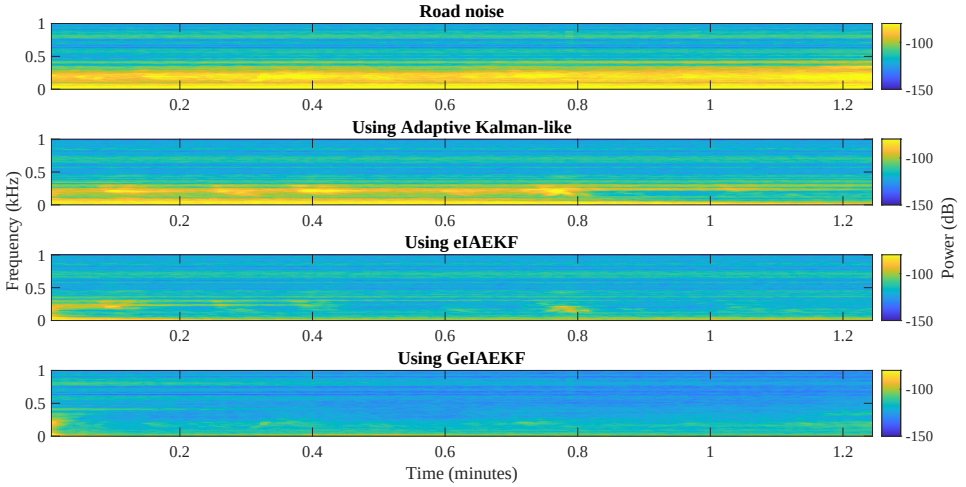


Figure 11: Spectrogram of the residual noise obtained at the *em*1 sensor using the Adaptive Kalman-like, the eIAEKF, and the GeIAEKF algorithms in the SISO ANC system canceling non-stationary road noise.

and stable tracking under gradual acoustic-path variations.

For non-stationary road noise, the primary noise spectrum is continuously changing. Fig 11 shows that GeIAEKF achieves a mean logarithmic spectral level of -121.08 dB, approximately 4.05 dB and 8.74 dB lower than eIAEKF and the adaptive Kalman-like algorithm, respectively.

These results demonstrate that GeIAEKF achieves better initial convergence than eIAEKF because its process-noise covariance is adapted online. In contrast, in eIAEKF it is fixed. GeIAEKF has improved residual noise suppression while maintaining stable tracking compared with the adaptive Kalman-like filter, because its covariance adaptation and Kalman gain are directly dependent on the measured error signal. Furthermore, online process-noise covariance adaptation reduces dependence on manually tuned covariance parameters compared with the original eIAEKF.

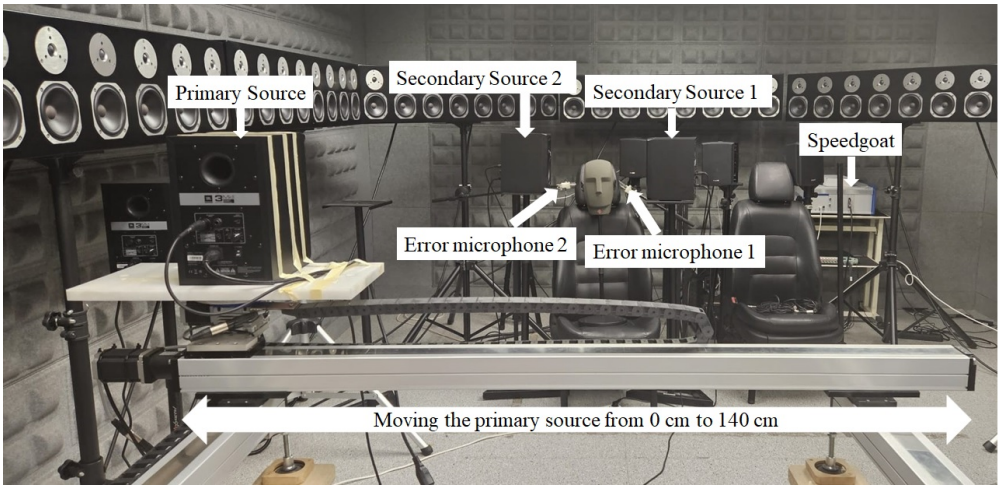


Figure 12: A MIMO experimental setup for the active noise control tests.

4. Experimental Results

4.1. MIMO Experimental Setup

This section presents real-time experiments evaluating the proposed GeIAEKF algorithm against the NLMS and SKF for MIMO ANC systems.

A real-time MIMO 1x2x2 ANC system was implemented using the Speedgoat IO104 platform in conjunction with MATLAB/Simulink, as detailed in (Islam, Garzon, Ferrer, Diego, Fuster and Gonzalez, 2024; Islam, Garcia, Garzon, Ferrer, de Diego, Fuster and Gonzalez, 2025) This experimental setup involved several key components: a loudspeaker generating white noise serving as the primary source, two error microphones positioned 2.35 meters (em1) and 2.25 meters (em2) away to capture residual noise, and two secondary speakers located 2.36 meters (ss1) and 2.24 meters (ss2) from the initial primary source location to facilitate active noise cancellation The primary noise source is placed on a platform that can move it with a 10 cm/s speed The dimensions of the room used for the experiment were 9.13 meters in length, 4.48 meters in width, and 2.637 meters in height, providing a realistic acoustic environment for the testing, as shown in Figure 12 The room exhibited a reverberation time of 0.242 seconds, indicating how sound persists in the space, with a decay rate of 60 dB.

The adaptive filter length was $N = 200$, secondary path length $M = 70$, and sampling frequency $f_s = 1000$ Hz.

Throughout the experiments, the NLMS algorithm uses an initial step size $\mu = 0.001$. The SKF control uses $\mathbf{Q} = (4 \times 10^{-10})\mathbf{I}_N$, $\mathbf{R} = (1 \times 10^{-6})\mathbf{I}_K$, $\mathbf{P}(0) = (5 \times 10^{-6})\mathbf{I}_N$, and $\mathbf{w}(0) = \mathbf{0}$. The process and measurement noise covariance values are manually tuned by trial and error While the initial state error covariance matrix $\mathbf{P}(0)$ is set by putting the mean of the secondary path energy on the diagonal The GeIAEKF control does not require initialization of the process and measurement noise covariance; we set the initial values $\mathbf{Q}(0) = \mathbf{0}$, $\mathbf{R}(0) = \mathbf{I}_K$, $\mathbf{Q} = (4 \times 10^{-10})\mathbf{I}_N$, $\epsilon = 0.000001$, and $\alpha = 0.9999$.

4.1.1. Response to a Static Noise Source

In this scenario, the primary noise source is kept fixed at its initial position.

The time-domain results in Fig. 13 show that NLMS converges slowly and exhibits significant steady-state fluctuations. The properly tuned SKF achieves rapid convergence and maintains stable residual levels. However, tuning the SKF required extensive trial and error, which was very time-consuming. The proposed GeIAEKF algorithm performs much better than the NLMS algorithm and closely matches the performance of the SKF. The proposed GeIAEKF algorithm required some time during the transient period to adapt its parameters.

The spectrogram analysis in Fig. 14 shows that SKF achieves the strongest attenuation across the frequency range. The GeIAEKF closely follows, typically within 3–6 dB, while NLMS shows significantly higher residual energy.

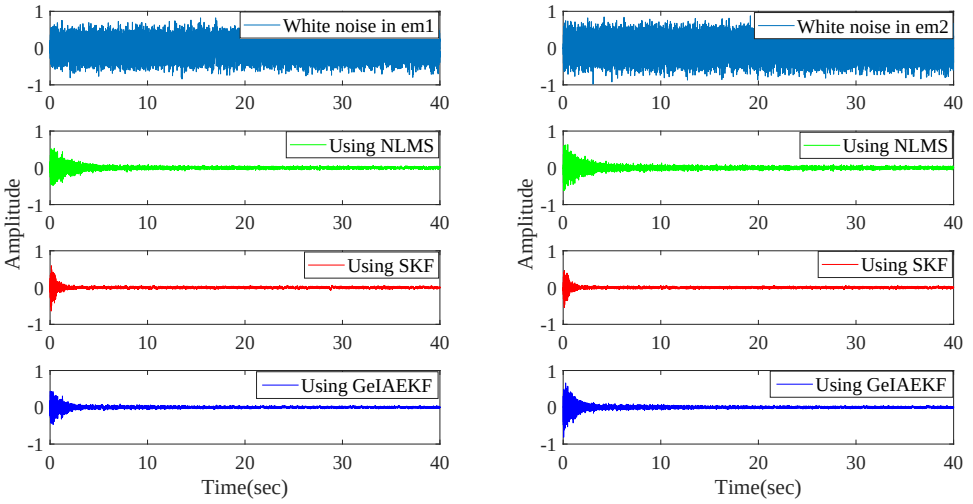


Figure 13: Noise and error signals of a static real-time MIMO ANC system Parameter setting: $mu = 0.001$, $\mathbf{Q} = (1 \times 10^{-10})\mathbf{I}_{LIJ}$, $\mathbf{R}(0) = \mathbf{I}_K$, $\mathbf{P}(0) = (5 \times 10^{-6})\mathbf{I}_{LIJ}$, $\mathbf{w}(0) = 0$, $\epsilon = 0.00001$, and $\alpha = 0.9999$.

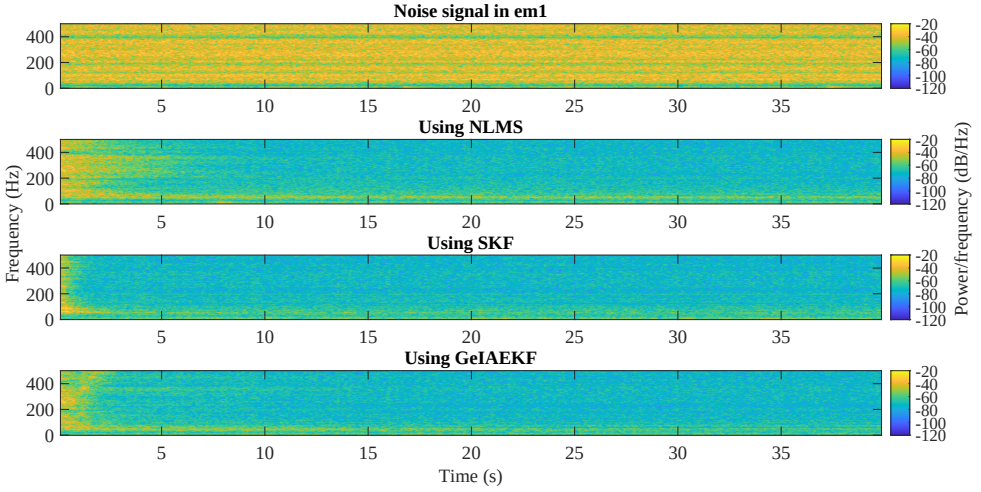


Figure 14: Spectrogram of the residual noise signals measured at the error microphones of a static real-time MIMO ANC system. Parameter setting: $mu = 0.001$, $\mathbf{Q} = (4 \times 10^{-10})\mathbf{I}_{LIJ}$, $\mathbf{R}(0) = \mathbf{I}_K$, $\mathbf{P}(0) = (5 \times 10^{-6})\mathbf{I}_{LIJ}$, $\mathbf{w}(0) = 0$, $\epsilon = 0.00001$, and $\alpha = 0.9999$.

4.1.2. Response to a Moving Noise Source

In this case, the primary noise source was moved from 0 cm to 140 cm and back at a speed of 10 cm/s. Under these dynamic conditions, the NLMS fails to maintain the attenuation achieved in the static case, whereas both the SKF and the GeIAEKF successfully track the changes.

The time-domain results in Fig. 15 show that the SKF achieves slightly better steady-state attenuation but is more sensitive to parameter tuning and to transient oscillations. The GeIAEKF maintains stable performance with reduced fluctuations.

The spectrogram analysis in Fig. 16 shows that the GeIAEKF achieves broadband noise reduction comparable to SKF while maintaining smoother adaptation behavior.

Overall, the results demonstrate that while the SKF achieves the highest absolute attenuation, the proposed GeIAEKF attains near-equivalent performance (within 2–5 dB) with fast convergence, improved transient stability, and

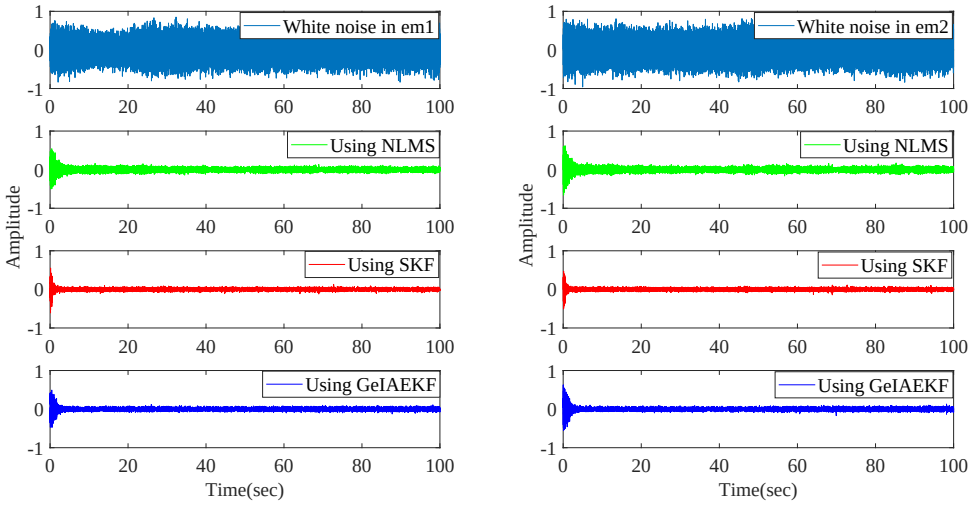


Figure 15: Noise and error signals of a real-time MIMO ANC system where the primary source is moving. Parameter setting: $mu = 0.001$, $\mathbf{Q} = (1 \times 10^{-10})\mathbf{I}_{LIJ}$, $\mathbf{R}(0) = \mathbf{I}_K$, $\mathbf{P}(0) = (5 \times 10^{-6})\mathbf{I}_{LIJ}$, $\mathbf{w}(0) = 0$, $\epsilon = 0.00001$, and $\alpha = 0.9999$.

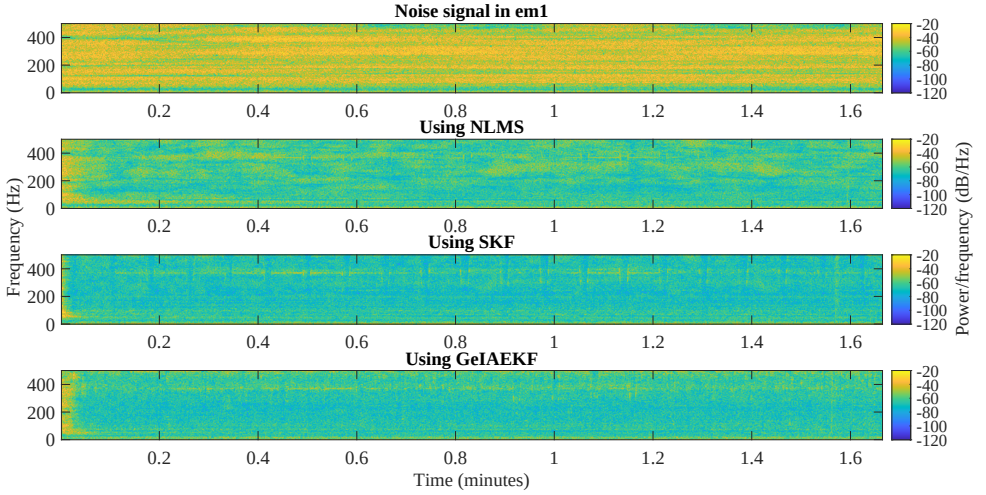


Figure 16: Spectrogram of the residual noise signals measured at the error microphone of a real-time MIMO ANC system where the primary source is moving. Parameter setting: $mu = 0.001$, $\mathbf{Q} = (1 \times 10^{-10})\mathbf{I}_{LIJ}$, $\mathbf{R}(0) = \mathbf{I}_K$, $\mathbf{P}(0) = (5 \times 10^{-6})\mathbf{I}_{LIJ}$, $\mathbf{w}(0) = 0$, $\epsilon = 0.00001$, and $\alpha = 0.9999$.

robustness since it does not require manual covariance tuning. Combined with its significantly lower computational complexity and ability to operate at higher sampling rates, the GeIAEKF provides a practical, robust, and scalable alternative for real-time ANC applications.

4.2. SISO Experimental Setup

A real-time SISO 1x1x1 ANC system was implemented using the Speedgoat IO104 platform in conjunction with MATLAB/Simulink. This experimental setup involved several key components: a loudspeaker generating white noise serving as the primary source, one error microphone positioned 2.05 meters (em1) away to capture residual noise, and one secondary speaker located 2.16 meters (ss1) from the initial primary source location to facilitate active noise cancellation, providing a realistic acoustic environment for testing, as shown in Figure 17. Throughout the experiments, the initializations described in Section 4.1 are used for the control algorithms.

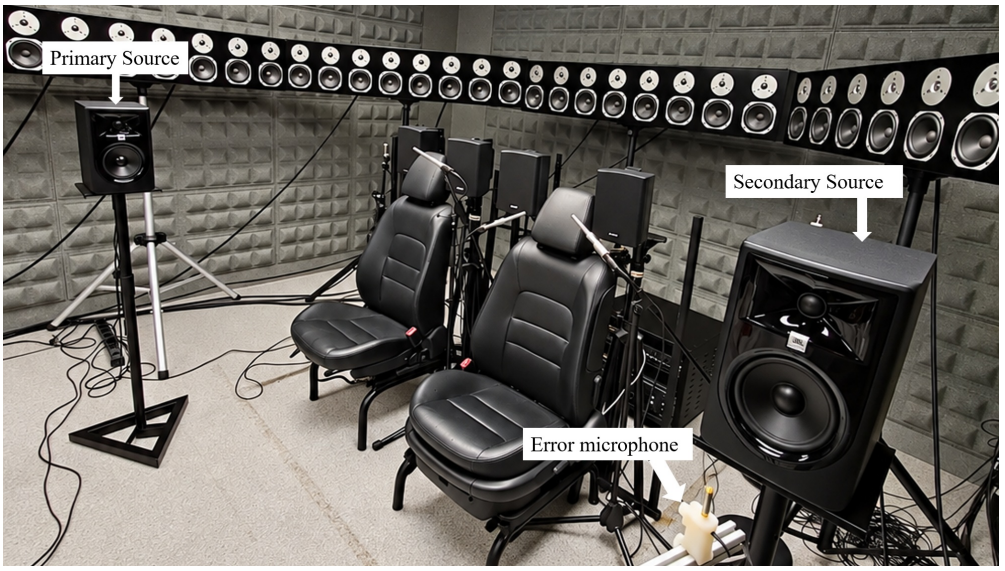


Figure 17: A SISO experimental setup for the active noise control tests.

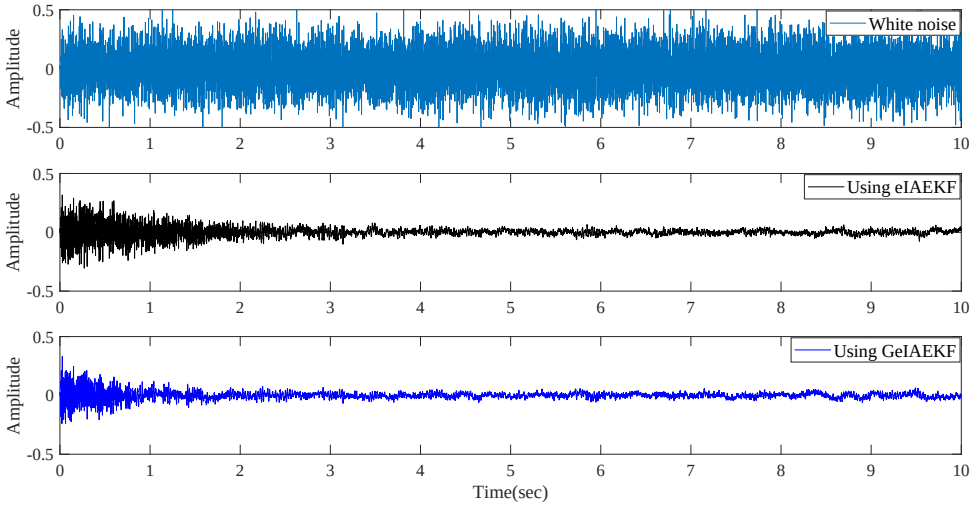


Figure 18: Noise and error signals of a static real-time SISO ANC system Parameter setting: $mu = 0.001$, $\mathbf{Q} = (1\times10^{-10})\mathbf{I}_{LIJ}$, $\mathbf{R}(0) = \mathbf{I}_K$, $\mathbf{P}(0) = (5\times10^{-6})\mathbf{I}_{LIJ}$, $\mathbf{w}(0) = 0$, $\epsilon = 0.00001$, and $\alpha = 0.9999$.

4.2.1. Response of SISO Static System

The real-time experiments compare the proposed GeIAEKF with the previously developed eIAEKF. The adaptive Kalman-like algorithm was not implemented on the Speedgoat platform because its online adaptation of the state-transition matrix $\mathbf{F}$ requires the full Kalman prediction model, which exceeds the platform's implementation constraints.

Figure 18 presents the measured noise signals and residual errors in a SISO ANC system. The GeIAEKF achieves faster convergence with a noise attenuation of 14.59 dB, compared with 12.30 dB for eIAEKF. Thus, GeIAEKF provides approximately 2.30 dB additional attenuation.

The corresponding spectrograms in Fig. 19 further demonstrate the effectiveness of both algorithms. GeIAEKF

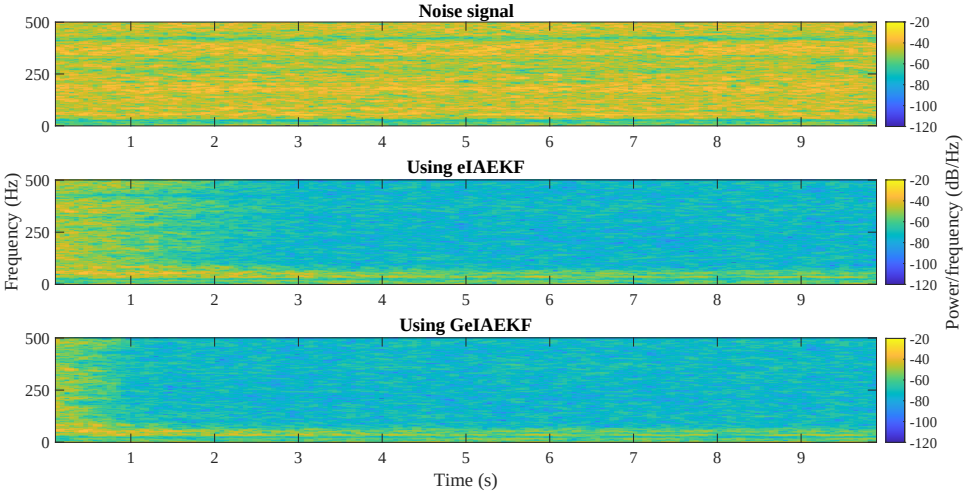


Figure 19: Spectrogram of the residual noise signals measured at the error microphones of a static real-time SISO ANC system. Parameter setting: $mu = 0.001$, $\mathbf{Q} = (4 \times 10^{-10})\mathbf{I}_{LIJ}$, $\mathbf{R}(0) = \mathbf{I}_K$, $\mathbf{P}(0) = (5 \times 10^{-6})\mathbf{I}_{LIJ}$, $\mathbf{w}(0) = 0$, $\epsilon = 0.00001$, and $\alpha = 0.9999$.

achieves a lower mean logarithmic spectral level (-69.79 dB) than eIAEKF (-67.93 dB), indicating improved residual-noise suppression.

These real-time results demonstrate that the proposed GeIAEKF provides improved noise attenuation compared with eIAEKF while remaining suitable for implementation on the Speedgoat platform.

5. Conclusion

This paper presented a generalized error innovation-based adaptive estimation Kalman filter (GeIAEKF) for filtered-x MIMO active noise control. Unlike conventional Kalman-based ANC approaches that rely on fixed or manually tuned covariance matrices, the proposed method directly uses the measured error signal to adapt the innovation and process-noise covariances online. This formulation ensures consistency with the ANC objective and controller robustness while significantly reducing computational complexity.

Simulation results demonstrated that while the SKF achieves the highest steady-state attenuation, the proposed GeIAEKF attains comparable performance within approximately 2–5 dB across stationary, non-stationary, and structural-variation scenarios. In addition, the GeIAEKF consistently exhibits fast convergence and improved transient behavior, particularly under abrupt spectral changes and noise-source relocation, due to covariance-matrix adaptation.

When comparing adaptive Kalman filter algorithms under gradual and continuous system dynamics, the GeIAEKF consistently achieves faster convergence and superior transient behavior compared to the eIAEKF and the adaptive Kalman-like algorithms.

Overall, the proposed GeIAEKF bridges the gap between high-performance state-estimation-based ANC and real-time implementability. It provides a fully adaptive solution that maintains near-optimal noise attenuation without requiring prior parameter tuning, making it a strong candidate for deployment in embedded and industrial active noise control applications. Future work will focus on extending the proposed framework to larger-scale MIMO systems and incorporating secondary-path adaptation for fully self-configuring ANC systems in highly dynamic environments.

Declaration of competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRediT authorship contribution statement

Tahmida Islam: Writing – review and editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.. **Miguel Ferrer:** Writing – review and editing, Validation, Supervision, Resources, Formal analysis, Data curation, Conceptualization.. **Maria de-Diego:** Writing – review and editing, Writing – original draft, Validation, Supervision, Project administration, Funding acquisition, Conceptualization.. **Alberto Gonzalez:** Writing – review and editing, Writing – original draft, Validation, Supervision, Project administration, Funding acquisition, Conceptualization..

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